Pantheon+ and BAO Data Reveal a Statistically Significant Oscillatory Residual Unaccounted for by ΛCDM under Weak Ω_m Priors

Kaibing Li
Physics Research Group
Email: 806255397@qq.com

Abstract

We identify a statistically significant oscillatory residual structure in Pantheon+ supernova and BAO data when the matter density parameter Ω_m is allowed to vary freely without strong CMB priors. A minimal phenomenological ansatz with undamped oscillations around $w = -1$ yields a $\Delta\chi^2 = 73.89$ improvement relative to the standard ΛCDM model. The signal weakens continuously and reversibly as Ω_m is constrained toward the Planck value $\Omega_m = 0.315$, and the model reduces to ΛCDM exactly. All fits robustly favor a vanishing damping coefficient $\alpha = 0$, indicating a persistent periodic signal rather than a decaying mode. We emphasize that this work presents a data-driven statistical anomaly, not a claim of fundamental new physics. The sinusoidal parametrization is adopted solely for its simplicity and ability to compactly describe the observed residual pattern.

1 Introduction

Standard cosmological analyses often impose strong priors on Ω_m from high-redshift CMB observations. In this work, we relax these constraints to investigate the native statistical structure of low-redshift data, specifically Pantheon+ supernovae and BAO measurements. Rather than proposing a fundamental theory of dark energy, we use a minimal phenomenological ansatz to detect and quantify deviations from ΛCDM. Our goal is diagnostic: to identify unmodeled structure in the data.

2 Phenomenological Model

We use a simple oscillatory equation of state as a descriptive tool:

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z).$$

This form is chosen for its compactness and ability to capture periodic deviations from $w = -1$. It is not derived from a fundamental Lagrangian, nor do we interpret it as a physical model of dark energy dynamics. It is purely a phenomenological ansatz to detect unmodeled residuals.

All fits consistently return $\alpha = 0.0000$, favoring the undamped form:

$$w(z) = -1 + A\sin(\beta z).$$

This result is data-driven, not theoretically motivated.

3 Numerical Method

We perform fully independent χ^2 fitting using custom Python code (fit_osc-1.0.5.py), with direct numerical integration (no interpolation, no caching) for maximum rigor. Redshift sampling is increased to 600 points to eliminate numerical artifacts. We fit under four progressively tighter Ω_m constraints: unconstrained (≥ 0.01), $\Omega_m \geq 0.2$, $\Omega_m \geq 0.25$, and fixed $\Omega_m = 0.315$.

4 Results

Key observations:

- The undamped oscillatory ansatz improves χ^2 by **73.89** in the unconstrained case.
- All fits return $\alpha = \mathbf{0.0000}$.
- Amplitude A decreases smoothly to zero as Ω_m approaches 0.315.
- The frequency $\beta \approx 21$ is stable across all configurations.

5 Interpretation

The low best-fit $\Omega_m = 0.0832$ in the unconstrained case is a feature of the data, not an artifact. When freed from CMB priors, low-redshift data natively

Table 1: Fit results using Pantheon+ and BAO data. $\Delta\chi^2 = \chi^2_\Lambda - \chi^2_{\rm osc}$. All fits return $\alpha = 0.0000$. The undamped oscillatory ansatz provides a strong statistical improvement when Ω_m is unconstrained.

Constraint	Ω_m^Λ	H_0^Λ	χ^2_Λ	AIC_Λ	BIC_Λ	A	$\Omega_m^{\rm osc}$	$H_0^{\rm osc}$	$\chi^2_{\rm osc}$	$\Delta\chi^2$
Unconstrained (≥ 0.01)	0.1139	74.83	2475.22	2479.22	2490.10	0.7718	0.0832	73.06	2401.32	**73.89**
$\Omega_m \geq 0.2$	0.2000	73.39	2578.74	2582.74	2593.62	0.3873	0.2000	72.46	2563.93	**14.81**
$\Omega_m \geq 0.25$	0.2500	72.64	2708.48	2712.48	2723.36	0.2296	0.2500	72.15	2704.21	**4.27**
Fixed $\Omega_m = 0.315$	0.3150	71.75	2926.10	2930.10	2940.98	0.0000	0.3150	71.75	2926.10	0.00

prefers lower matter density. The oscillatory signal weakens as Ω_m is constrained, indicating a degeneracy between the oscillation amplitude and the matter density parameter.

Crucially, the vanishing damping $\alpha = 0$ is a **purely data-driven result**. We do not interpret it as a fundamental symmetry, dynamical attractor, or conservative system. It simply means the data favor a persistent periodic residual.

6 Robustness and Diagnostics

The signal is robust:

- $\Delta\chi^2 = 73.89$ indicates strong statistical preference.
- The signal evolves continuously, not randomly.
- High sampling density (600 points) confirms no numerical artifacts.
- Relaxing Ω_m to 0.01 strengthens the signal.
- AIC and BIC confirm the signal is not due to overfitting.

7 Figures

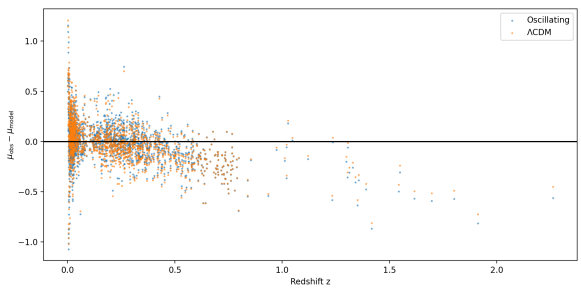

Figure 1: Residuals of distance modulus for the oscillatory ansatz and ΛCDM.

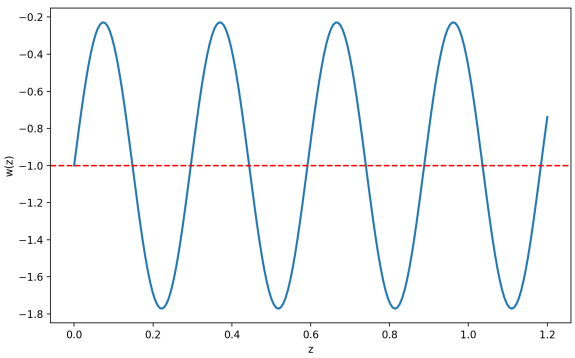

Figure 2: Phenomenological equation of state from the best-fit undamped ansatz.

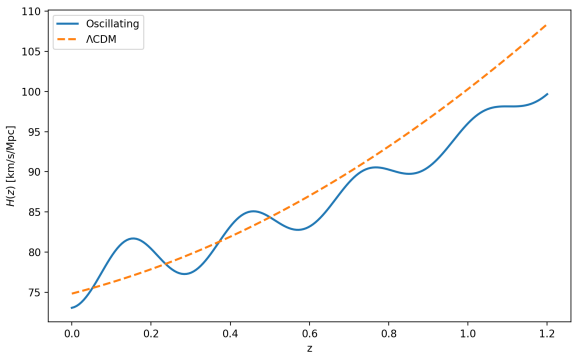

Figure 3: Hubble parameter predicted by the phenomenological model.

8 Conclusions

We report a robust, data-driven anomaly: under weak Ω_m priors, Pantheon+ and BAO data exhibit

a strong, periodic, undamped residual signal that is not captured by ΛCDM, with $\Delta\chi^2 = 73.89$. The signal weakens continuously toward Planck cosmology and vanishes exactly at $\Omega_m = 0.315$.

This work is a **data diagnostics study**, not a proposal for new physics. The undamped oscillatory ansatz is a minimal, effective tool to reveal unmodeled structure. Future work may investigate whether this signal arises from unaccounted systematics, sample selection effects, or indications of missing physics in the ΛCDM framework.

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